INDIRECT ADDITIVE MANUFACTURING OF A LUNAR REGOLITH SIMULANT

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1 Introduction

Ceramic part printing generally requires, in most processes, passing through a **"green part"** stage composed of a mixture of a **ceramic powder** and a **polymer binder**. These sacrificial binders are removed and lost during thermal treatments. To mitigate the environmental impact of gases released during debinding, this project aims to valorize these gaseous emissions as a feedstock for bacterial growth, enabling binder regeneration and ultimately closing the recycling loop.



Fig1: Illustration of the lunar surface during futur lunar exploration (by ESA)

In a space context, future lunar missions will require the ability to **manufacture** various objects on site. Transporting resources from Earth will no longer be feasible, so the polymer binder must be **produced** and **recycled locally**. The ceramic fraction, however, is abundant as it can be directly sourced from lunar regolith.

2 Process Flow Diagram

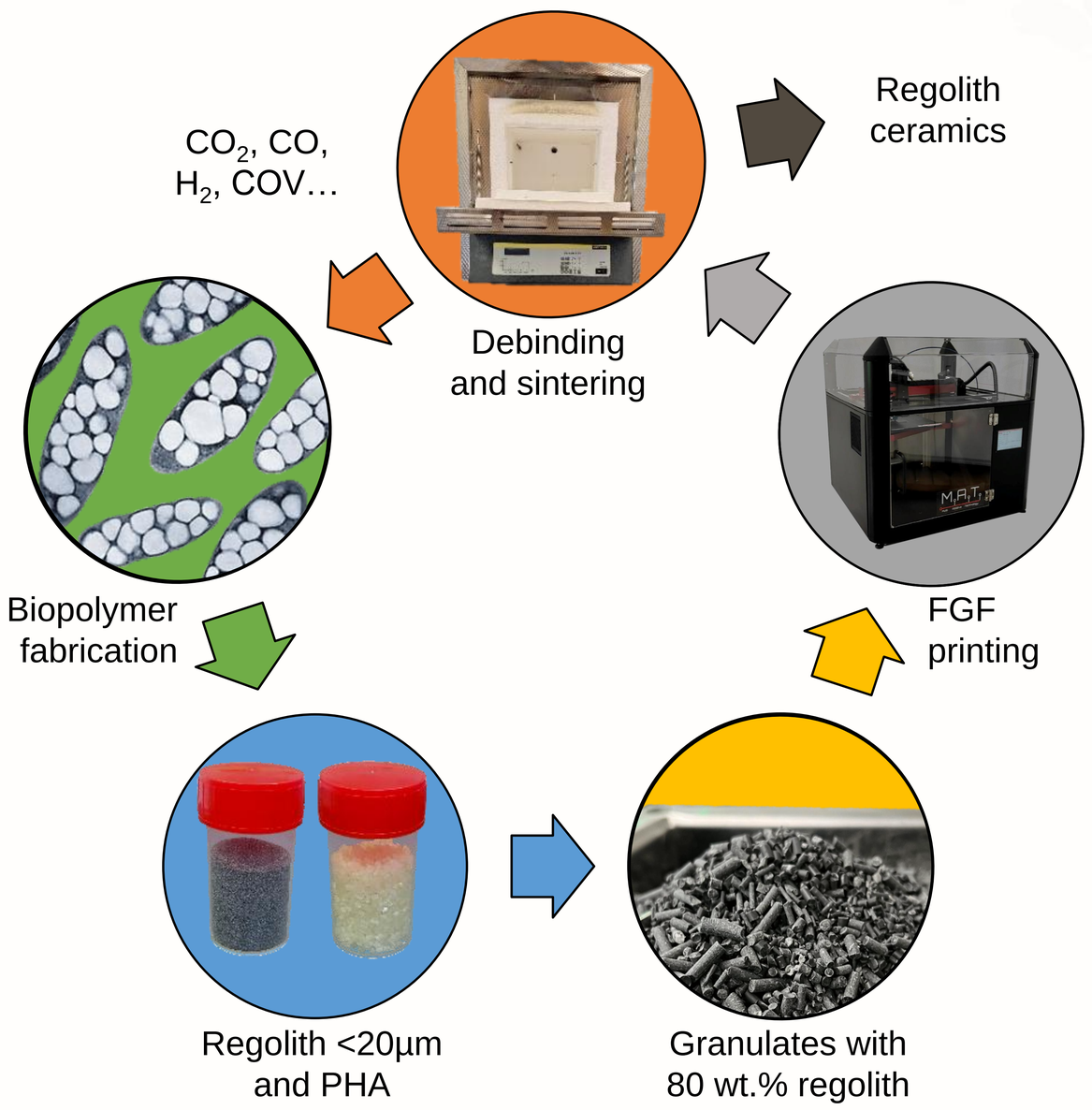


Fig 2: Close-loop regolith ceramics fabrication with binder recycling

3 Methodology

- Recovery of fine lunar regolith powder (<20 µm) from the sieving process.
- Mixing of the regolith powder with an on-site-synthesized PHA biopolymer.
- Production of regolith-PHA granules.
- Fabrication of the green part using the Fused Granulate Fabrication (FGF) process.
- Thermal degradation of the PHA binder in the green part and reuse of the generated organic gases as a feedstock for the bioreactor.
- Sintering of the brown part to obtain a ceramic component made from lunar regolith.

4 Thermal debinding (Macro TGA)

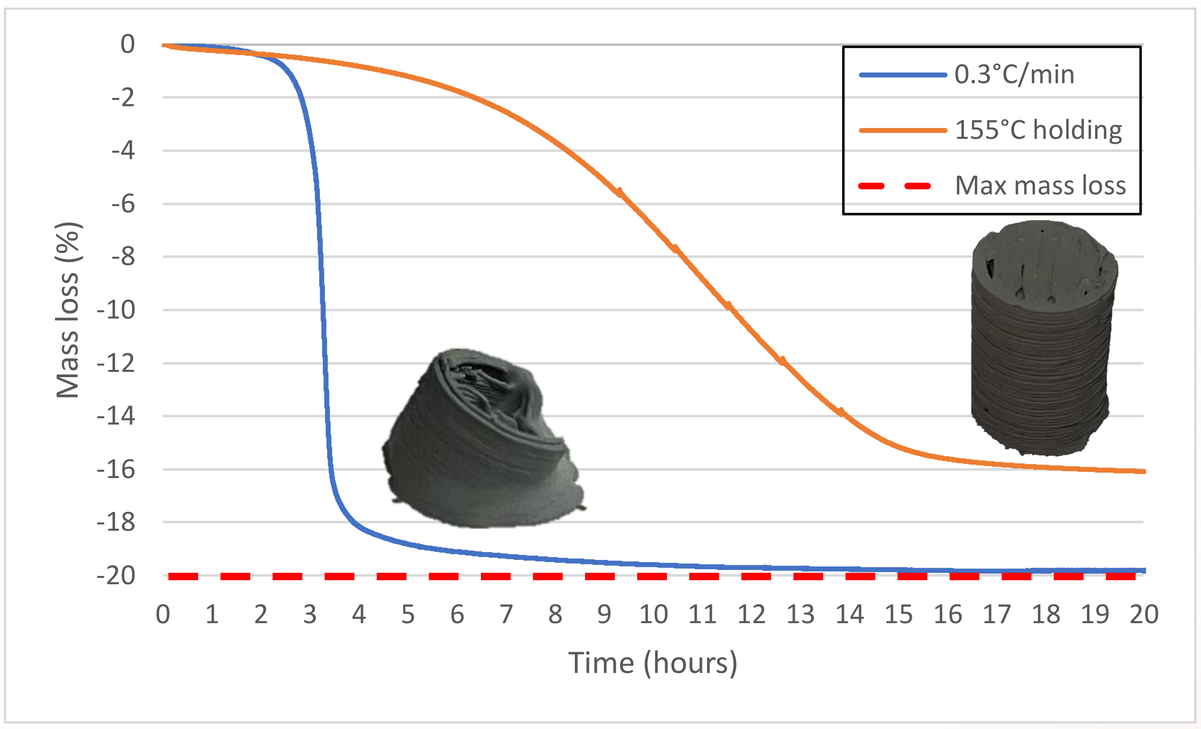


*Fig 4: Mass loss monitored for **0,3 °C/min heat up** Vs. **24h dwell at 155 °C***

- **Conventional thermal debinding**

When following a conventional heating ramp of 0.3 °C /min, the mass-loss kinetics become too rapid, causing the sample to lose its structural integrity within a few minutes.

- **Isothermal debinding below the melting temperature (Tm)**

By maintaining the temperature a few degrees below Tm, the polymer fraction that has melted undergoes random chain scission and subsequent volatilization. This approach enables up to 80% debinding after 24 hours at 155 °C. It promotes the controlled release of low-molecular-weight degradation products, resulting in a slower and more controlled mass-loss rate.

5 Post processing steps

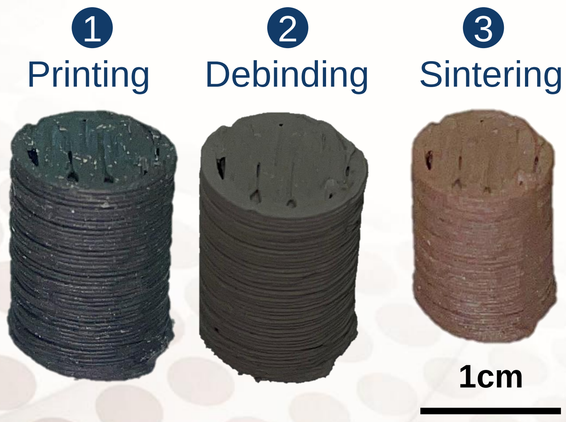


Fig 3: Printing, thermal debinding and sintering steps of a sample

1 → 2: Printing – Debinding
•Controlled removal of the organic binder.
•Controlled mass loss to prevent defects (bubbles, cracks, deformation).

2 → 3: Debinding – Sintering
•Shrinkage due to densification and pore closure.
•Iron oxidation: **Fe^{2+} → Fe^{3+}**.